

SEDS 537 – Machine Learning | Term Project Proposal

Project Title: Multilingual Toxic Comment Classification



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My proposal: Multilingual Toxic Comment Classification

Problem Description: Detecting toxic comments across multiple languages is challenging. This project aims to build an efficient multilingual toxic comment classifier using parameter-efficient fine-tuning.

Proposed Approach: Fine-tune mBERT using adapter modules instead of updating all parameters. This reduces training cost while maintaining cross-lingual performance.

Baseline Methods:

1. **TF-IDF + Logistic Regression:** A simple and interpretable baseline for text classification. I will use TF-IDF features with Logistic Regression to establish a minimum performance benchmark.
2. **TF-IDF + Naive Bayes:** A probabilistic classifier well-suited for text data. I will use it to compare other classical methods under the same TF-IDF representation.
3. **TF-IDF + SVM:** Known to perform well on high-dimensional text features. I will use it as the strongest classical ML baseline.
4. **LSTM:** A recurrent neural network that captures word order and sequential patterns, unlike TF-IDF-based methods. I will train it with multilingual word embeddings to test a simple deep learning approach.
5. **mBERT (full fine-tuning):** A pretrained multilingual transformer covering 104 languages. I will fine-tune all parameters on English data and test cross-lingual transfer, serving as the strongest deep learning baseline to compare against my proposed adapter method.

Dataset: Jigsaw Multilingual Toxic Comment Classification (Kaggle). English training data, multilingual test data (Turkish, Spanish, Italian, etc.)

Evaluation Strategy: AUC-ROC, F1-score, per-language accuracy, inference time comparison, ablation study on adapters vs full fine-tuning.

Expected Challenges: Class imbalance in toxicity labels, cross-lingual generalization, and ensuring adapters capture language-specific toxic patterns effectively.